

Nonlinear Craig Interpolant Generation over Unbounded Domains

Separating Semialgebraic Sets

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Craig Interpolation

Theorem (Craig, 1957)

Let ϕ and ψ be two formulas in a first-order theory $\mathcal{T}$. If $\phi \models \psi$, then there exists a formula ρ such that:

- ① $\phi \models \rho$ and $\rho \models \psi$,
- ② All non-logical symbols in ρ occur in ϕ and ψ .

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An important **local reasoning technique** in formal methods:

- theorem proving: Krajíček, 1997; Pudlák, 1997
- model checking: K. McMillan, 2003
- abstraction refinement: Henzinger et al., 2004; K. McMillan, 2005
- program verification: Jung et al., 2011; S. Lin et al., 2017
- ...

Synthesizing Interpolants

For interpolation-based techniques, the **efficient synthesis** of Craig interpolants is crucial. Researchers have developed algorithms for different theories:

- Linear arithmetic Henzinger et al., 2004
- Logic with arrays Hoenicke et al., 2018; K. L. McMillan, 2008
- Logic with sets Kapur et al., 2006
- Equality logic with uninterpreted functions (EUF) K. McMillan, 2005; Cimatti et al., 2008
- Combinations of various theories Yorsh et al., 2005; Kovács et al., 2009; Rybalchenko et al., 2010.

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- **Nonlinear Real Arithmetic (NRA)**
 - by “nonlinear” we mean polynomial

Problem Formulation

- W.o.l.g, let ϕ and ψ be conjunctions of polynomial inequalities:

$$\phi(\mathbf{x}, \mathbf{y}) = \bigwedge_{i=1}^n f_i(\mathbf{x}, \mathbf{y}) \geq 0, \quad \psi(\mathbf{x}, \mathbf{z}) = \bigwedge_{j=1}^m g_j(\mathbf{x}, \mathbf{z}) \geq 0.$$

where f_i, g_j are polynomials.

- Problem** Assume that $\phi(\mathbf{x}, \mathbf{y}) \models \neg\psi(\mathbf{x}, \mathbf{z})$, find a function $h(\mathbf{x})$ such that the formula $h(\mathbf{x}) > 0$ is an interpolant for ϕ and $\neg\psi$, i.e.,

$$\phi(\mathbf{x}, \mathbf{y}) \models h(\mathbf{x}) > 0 \text{ and } \psi(\mathbf{x}, \mathbf{z}) \models h(\mathbf{x}) < 0.$$

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- In other words, $h(\mathbf{x}) = 0$ separates the two (basic) semialgebraic sets S_ϕ and S_ψ .

A Summary of Related Works

Reference	Restriction	Technique	Complete	Solver
Dai et al., 2013	no unshared variables, Archimedean cond.	Stengle's Thm	✓	SDP
Gan et al., 2016	concave quadratic	Motzkin's transposition Thm.	✓	SDP
Gan et al., 2020	Archimedean cond.	Putinar's Thm.	✓	SDP
This work	general	Putinar's Thm., homogenization	✓	SDP
Jovanović et al., 2021	general	Quantifier Elimination (CAD)	✓	MCSat
Chen et al., 2019	general	CEGIS	✗	SVM
W. Lin et al., 2022	general	CEGIS	✗	NN
Gao et al., 2016	general	Linearization, ICP	✗	δ -Real
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What is the Archimedean condition and how we resolve it?

Putinar's Positivstellensatz

- A polynomial $\sigma(\mathbf{x})$ is **Sum-of-Squares** (SOS) if it can be written as

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- $\Sigma[\mathbf{x}]$ denotes the set of SOS polynomials.
- A polynomial $f(\mathbf{x})$ is SOS iff the **Gram matrix** of f is PSD.

$$\begin{aligned} f(x_1, x_2) &= 2x_1^4 + 2x_1^3x_2 - x_1^2x_2^2 + 5x_2^4 \\ &= \underbrace{\begin{pmatrix} x_1^2 \\ x_2^2 \\ x_1x_2 \end{pmatrix}}_{\text{basis}}' \underbrace{\begin{pmatrix} 2 & -3 & 1 \\ -3 & 5 & 0 \\ 1 & 0 & 5 \end{pmatrix}}_{\text{Gram matrix}} \underbrace{\begin{pmatrix} x_1^2 \\ x_2^2 \\ x_1x_2 \end{pmatrix}}_{\text{basis}} \\ &= \frac{1}{2}(2x_1^2 - 3x_2^3 + x_1x_2)^2 + \frac{1}{2}(x_2^2 + 3x_1x_2)^2 \end{aligned}$$

Putinar's Positivstellensatz

- Let $S = \{x \mid p_i(x) \geq 0, i = 1, \dots, m\}$.
- **Quadratic Module**

$$\mathbf{QM}(S) \triangleq \left\{ \sigma_0 + \sum_{i=1}^m \sigma_i p_i \mid \sigma_i \in \Sigma[\mathbf{x}] \text{ for } 1 \leq i \leq m \right\}$$

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- A quadratic module $\mathbf{QM}$ is **Archimedean** if $N - \|x\|^2 \in \mathbf{QM}$ for some constant $N \in \mathbb{N}$.
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Theorem (Putinar's Positivstellensatz)

Assume the Archimedean condition holds. If $f \in \mathbb{R}[x]$ is *strictly positive* on S , then $f \in \mathbf{QM}(S)$, i.e.,

$$f(x) = \sigma_0(x) + \sum_{i=1}^m p_i(x) \cdot \sigma_i(x), \quad \sigma_i(x) \in \Sigma[x].$$

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- Let $p(\mathbf{x})$ be a polynomial of degree d , its homogenization is

$$\tilde{p}(\tilde{\mathbf{x}}) \triangleq x_0^d \cdot p\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right).$$

$$p(x_1, x_2) = x_1^2 + x_2 + 1 \implies \tilde{p}(x_0, x_1, x_2) = x_1^2 + x_2 x_0 + x_0^2.$$

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- Let $S = \{\mathbf{x} \mid p_1(\mathbf{x}) \geq 0, \dots, p_m(\mathbf{x}) \geq 0\}$, define

$$\tilde{S} \triangleq \{\tilde{\mathbf{x}} \in \mathbb{R}^{n+1} \mid \tilde{p}_1(\tilde{\mathbf{x}}) \geq 0, \dots, \tilde{p}_m(\tilde{\mathbf{x}}) \geq 0, \underbrace{x_0 \geq 0, \|\tilde{\mathbf{x}}\|^2 = 1}_{\text{half unit sphere in } \mathbb{R}^{n+1}}\}.$$

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There exists a **one-to-one mapping** between S and $\tilde{S} \setminus \{x_0 = 0\}$:

$$\mathbf{x} \in S \iff \left(\frac{1}{\sqrt{1 + \|\mathbf{x}\|^2}}, \frac{x_1}{\sqrt{1 + \|\mathbf{x}\|^2}}, \dots, \frac{x_n}{\sqrt{1 + \|\mathbf{x}\|^2}} \right) \in \tilde{S} \setminus \{x_0 = 0\}.$$

Or equivalently,

$$(x_0, \mathbf{x}) \in \tilde{S} \setminus \{x_0 = 0\} \iff \left(\frac{x_1}{\sqrt{1 - \|\mathbf{x}\|^2}}, \dots, \frac{x_n}{\sqrt{1 - \|\mathbf{x}\|^2}} \right) \in S$$

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Points with $x_0 = 0$ correspond to **points at infinity** in $\mathbb{R}^n$.

Polynomial Homogenization

Definition (closed at infinity (Nie, 2012))

A basic semialgebraic set S is closed at ∞ if

$$cl(\tilde{S} \setminus \{x_0 = 0\}) = \tilde{S}.$$

- Note

- 1 Being closed at infinity is a generic property for semialgebraic sets.
- 2 Depends on description polynomials.

$$S_1 = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 - x_2^2 \geq 0\} \text{ not closed at } \infty$$

$$S_2 = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 - x_2^2 \geq 0, \underbrace{x_1 \geq 0}_{\text{redundant}}\} \text{ closed at } \infty$$

Polynomial Homogenization

Theorem (Equivalence (Huang et al., 2023))

When S is closed at ∞ ,

$$f(\mathbf{x}) \geq 0 \text{ over } S \iff \tilde{f}(\tilde{\mathbf{x}}) \geq 0 \text{ over } \tilde{S}$$

- The quadratic module $\mathbf{QM}(\tilde{S})$ is Archimedean.

The Main Theorem

$$S_\phi = \{\mathbf{x} \in \mathbb{R}^r \mid f_1(\mathbf{x}) \geq 0, \dots, f_m(\mathbf{x}) \geq 0\},$$

$$S_\psi = \{\mathbf{x} \in \mathbb{R}^r \mid g_1(\mathbf{x}) \geq 0, \dots, g_n(\mathbf{x}) \geq 0\}$$

Theorem (Main)

If $\tilde{S}_\phi \cap \tilde{S}_\psi = \emptyset$, then there exists a (semialgebraic) function $h(\mathbf{x})$ such that

$$\forall \mathbf{x} \in S_\phi. h(\mathbf{x}) > 0 \text{ and } \forall \mathbf{x} \in S_\psi. -h(\mathbf{x}) > 0.$$

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Proposition (Gan et al., 2020)

If S_ϕ and S_ψ are of the Archimedean form and $S_\phi \cap S_\psi = \emptyset$, then there exists a polynomial $h(\mathbf{x}) \in \mathbb{R}[\mathbf{x}]$ such that

$$\forall \mathbf{x} \in S_\phi. h(\mathbf{x}) > 0 \text{ and } \forall \mathbf{x} \in S_\psi. -h(\mathbf{x}) > 0.$$

Proof Idea

- Apply the proposition (Gan et al., 2020) from to $\tilde{S}_\phi$ and $\tilde{S}_\psi$ (bounded over $\mathbb{R}^{n+1}$), then there exists a polynomial interpolant $g \in \mathbb{R}[x_0, \mathbf{x}]$ for $\tilde{S}_\phi$ and $\tilde{S}_\psi$, i.e.,

$$\forall (x_0, \mathbf{x}) \in \tilde{S}_\phi. g(x_0, \mathbf{x}) > 0 \text{ and } \forall (x_0, \mathbf{x}) \in \tilde{S}_\psi. -g(x_0, \mathbf{x}) > 0.$$

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- Using the one-to-one mapping, we construct an interpolant function (not necessarily polynomial) for S_ϕ and S_ψ :

$$h(\mathbf{x}) \triangleq g\left(\frac{1}{\sqrt{\|\mathbf{x}\|^2 + 1}}, \frac{\mathbf{x}}{\sqrt{\|\mathbf{x}\|^2 + 1}}\right).$$

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- The results can be extended to allow unshared variables and disjunction in ϕ and ψ .

Polynomial and Semialgebraic Interpolants

- To remove all denominators, we introduce a multiplier

$$\begin{aligned} h(\mathbf{x}) &\triangleq (\sqrt{\|\mathbf{x}\|^2 + 1})^{\deg(g)} g\left(\frac{1}{\sqrt{\|\mathbf{x}\|^2 + 1}}, \frac{\mathbf{x}}{\sqrt{\|\mathbf{x}\|^2 + 1}}\right), \\ &= h_1(\mathbf{x}) + \sqrt{\|\mathbf{x}\|^2 + 1} \cdot h_2(\mathbf{x}), \end{aligned}$$

where higher-order terms of $\sqrt{\|\mathbf{x}\|^2 + 1}$ are absorbed into h_1 and h_2 .

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- We can consider two types of interpolants:
 - Polynomial interpolants:** $h_2(\mathbf{x})$ is the zero polynomial.
 - Semialgebraic interpolants:** the more general case (first proposed).

Complete Characterizations

- **Polynomial** $h(\mathbf{x}) \in \mathbb{R}[\mathbf{x}]$, apply Putinar's Positivstellensatz to

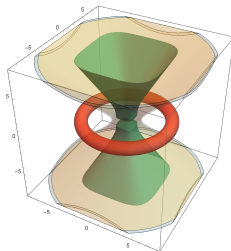
$$\tilde{h}(\mathbf{x}, x_0) > 0 \text{ over } \tilde{S}_\phi,$$

$$\tilde{h}(\mathbf{x}, x_0) < 0 \text{ over } \tilde{S}_\psi.$$

The resulting constraints can be solved by SDP.

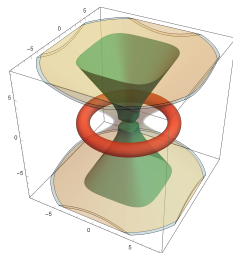
- **Semialgebraic** Similar, but we need to introduce a variable w to represent $\sqrt{\|\mathbf{x}\|^2 + 1}$, i.e., $h(\mathbf{x}, w) = h_1(\mathbf{x}) + w \cdot h_2(\mathbf{x}) \in \mathbb{R}[\mathbf{x}, w]$

An illustrative Example



$$\begin{aligned}
 \phi(x, y, z) &\triangleq 1 + 0.1z^4 - x^4 - y^4 \geq 0 \wedge 10z^4 - x^4 - y^4 \geq 0, \\
 \psi(x, y, z, r, R) &\triangleq 4R^2(x^2 + y^2) - (x^2 + y^2 + z^2 + R^2 - r^2)^2 \geq 0 \\
 &\quad \wedge 6 \geq R \geq 4 \wedge 1 \geq r \geq 0.5.
 \end{aligned}$$

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- polynomial:

$$h_p(x, y, z) = 1.0 - 0.35507338x^2 - 0.35507338y^2 + 0.45264895z^2.$$

- semialgebraic:

$$h_1(x, y, z) = 0.98004189 - 0.26291972x^2 - 0.26291978y^2 + 0.417581644z^2,$$

$$h_2(x, y, z) = 1.0 - 0.51670759x^2 - 0.51670759y^2 + 0.60569150z^2.$$

- The previous method (Gan et al., 2020) fails to find a polynomial interpolant of degree ≤ 4 .

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- The key is to **project** an unbounded region in $\mathbb{R}^n$ into a bounded region in $\mathbb{R}^{n+1}$ by using **homogenization**.

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Thanks! & Questions?



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